Infofys PYQ : Coding Questions

1. Minimum Coins Required

Given coins[] of size N and target value V, with an infinite supply of each denomination, find the minimum number of coins needed to make V. If it is not possible, return -1.

EXAMPLES

Input: coins[] = {25, 10, 5}, V = 30
Output: Minimum 2 coins required (25 + 5)

Input: coins[] = {9, 6, 5, 1}, V = 11
Output: Minimum 2 coins required (6 + 5)

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Explanation: Minimum Coins Required

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Explanation: Minimum Coins Required

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SOLUTION: JAVA

```java
static int findmincoins(int[] arr, int n, int x) {
    int min = Integer.MAX_VALUE;
    if (x == 0) {
        return 0;
    }
    for (int i = 0; i < n; i++) {
        if (arr[i] <= x) {
            int subresult = findmincoins(arr, n, x - arr[i]);
            if (subresult != Integer.MAX_VALUE && subresult + 1 < min) {
                min = subresult + 1;
            }
        }
    }
    return min;
}
```

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SOLUTION: CPP

```cpp
int findmincoins(vector<int>& arr, int n, int x) {
    int minCoins = INT_MAX;

    if (x == 0) {
        return 0;
    }

    for (int i = 0; i < n; i++) {
        if (arr[i] <= x) {
            int subresult = findmincoins(arr, n, x - arr[i]);

            if (subresult != INT_MAX && subresult + 1 < minCoins) {
                minCoins = subresult + 1;
            }
        }
    }

    return minCoins;
}
```

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2. Subset Sum

Given a set of non-negative integers and a target sum, determine whether any subset of the set adds up exactly to the target.

EXAMPLES

Input: set[] = {3, 34, 4, 12, 5, 2}, sum = 9
Output: True (subset {4, 5})

Input: set[] = {3, 34, 4, 12, 5, 2}, sum = 30
Output: False

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Explanation: Subset Sum

Infofys PYQ : psuedo codes

Campusmonk

Explanation: Subset Sum

Infofys PYQ : psuedo codes

JAVA

Campusmonk

```java
static boolean isSubsetSum(int[] arr, int n, int sum) {
    if (sum == 0) {
        return true;
    }

    if (n == 0) {
        return false;
    }

    if (arr[n - 1] > sum) {
        return isSubsetSum(arr, n - 1, sum);
    }

    return isSubsetSum(arr, n - 1, sum)
            || isSubsetSum(arr, n - 1, sum - arr[n - 1]);
}
```

Infofys PYQ : psuedo codes

C++

Campusmonk

```cpp
bool isSubsetSum(vector<int>& arr, int n, int sum) {
    if (sum == 0) {
        return true;
    }

    if (n == 0) {
        return false;
    }

    if (arr[n - 1] > sum) {
        return isSubsetSum(arr, n - 1, sum);
    }

    return isSubsetSum(arr, n - 1, sum)
            || isSubsetSum(arr, n - 1, sum - arr[n - 1]);
}
```

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3. Largest Number

Given an array of strings representing non-negative integers, arrange them so that concatenating them produces the largest possible number. Return the result as a string.

EXAMPLES

Input: n = 5, arr[] = {"3", "30", "34", "5", "9"}
Output: "9534330"

Input: n = 4, arr[] = {"54", "546", "548", "60"}
Output: "6054854654"

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Explanation: Largest Number

Infofys PYQ : Coding Questions

Explanation: Largest Number

Infofys PYQ : psuedo codes

JAVA

```java
public static String largestNumber(String[] arr) {
    Arrays.sort(arr, (a, b) -> (b + a).compareTo(a + b));

    if (arr[0].equals("0")) {
        return "0";
    }

    StringBuilder result = new StringBuilder();

    for (String num : arr) {
        result.append(num);
    }

    return result.toString();
}
```

Infofys PYQ : psuedo codes

C++

Campusmonk

```cpp
string largestNumber(vector<string>& arr)
{
    sort(arr.begin(), arr.end(), [](string& a, string& b)
    {
        return a + b > b + a;
    });

    if (arr[0] == "0")
        return "0";

    string result = "";
    for (string& num : arr)
        result += num;

    return result;
}
```